

HAT-P-54b

R-1302

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_211118 · created 2026-07-31T17:24:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459536.7835 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.046
Transit depth (Rp/Rs)^2	3.91 +/- 1.84 [%]
Orbital Inclination (inc)	83.67 +/- 2.17
Airmass coefficient 1 (a1)	1.218 +/- 0.055
Airmass coefficient 2 (a2)	-0.12 +/- 0.011
Scatter in the residuals of the lightcurve fit is	4.48 %
Best Comparison Star	None
Optimal Aperture	3.22
Optimal Annulus	18.95
Transit Duration (day)	0.042 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.198 $\pm$ 0.046 vs 0.1572 (deviation 0.8 σ)
Duration fit vs published	0.042 d vs 0.075 d (deviation 1.19 σ)
Mid-time O–C	5.6 min
Depth SNR	3.9
Residual scatter	4.48 %

Light curve

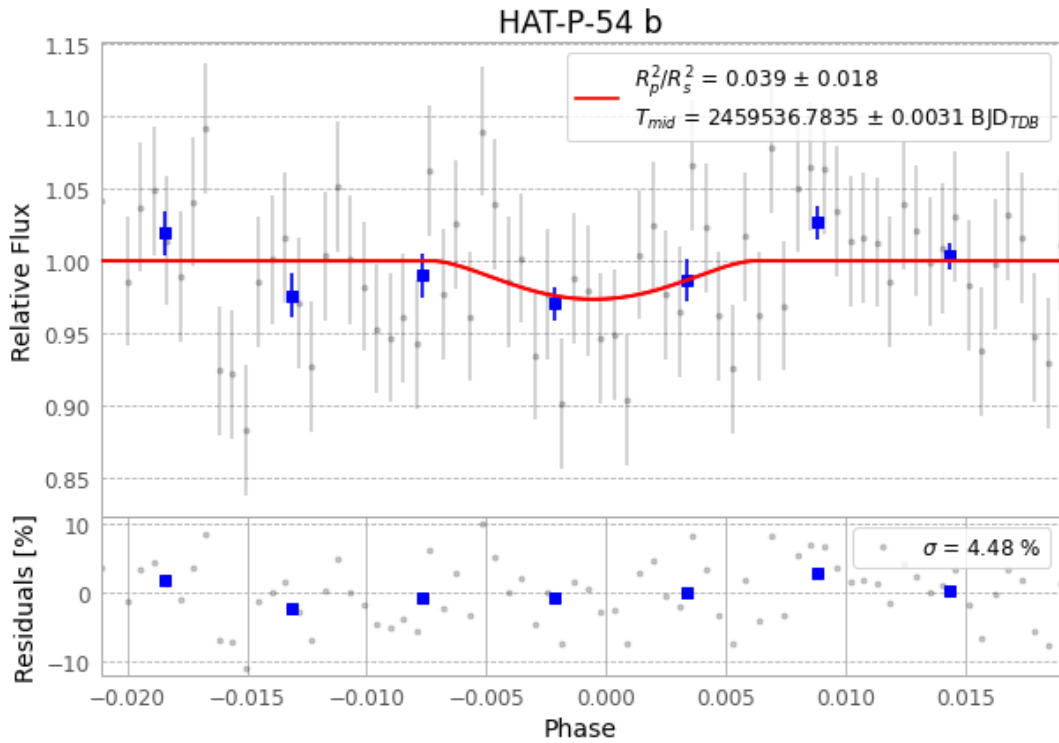


Figure 1 — FinalLightCurve_HAT-P-54 b_2021-11-17.png (SHA-256 16513e11672388d8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 47081fbd5a248a70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), HATP-54211118044510.FITS → HATP-54211118082412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-211118.log (bedd10f99f9e...), FinalLightCurve_HAT-P-54 b_2021-11-17.png (16513e116723...), FinalLightCurve_HAT-P-54 b_2021-11-17.csv (2a5c2dd8082e...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 17:24Z.